

OSLAB PRACTICAL-8 PROGRAMS & OUTPUT

Date: 01-08-26

Program-1 CODE & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_8$ cat program1.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *a, *b;

    a = malloc(3 * sizeof(int));
    printf("malloc: 3 integers allocated\n");

    b = calloc(3, sizeof(int));
    printf("calloc: 3 integers allocated\n");

    a = realloc(a, 5 * sizeof(int));
    printf("realloc: memory increased to 5 integers\n");

    free(a);
    free(b);
    printf("free: memory released\n");

    return 0;
}
sainath@Sainath:~/OSLAB/practical_8$ ./program1
malloc: 3 integers allocated
calloc: 3 integers allocated
realloc: memory increased to 5 integers
free: memory released
```

Program-2 CODE & OUTPUT:

```
Free memory released
sainath@Sainath:~/OSLAB/practical_8$ cat program2.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int *data = malloc(5 * sizeof(int));

    for (int i = 0; i < 5; i++)
        data[i] = 10;

    printf("Before fork: data[0] = %d\n", data[0]);

    pid_t pid = fork();

    if (pid == 0) {
        // Child
        printf("Child before modification: data[0] = %d\n", data[0]);

        data[0] = 99;

        printf("Child after modification: data[0] = %d\n", data[0]);

        free(data);
        exit(0);
    }
    else {
        // Parent
        wait(NULL);

        printf("Parent after child modification: data[0] = %d\n",
            data[0]);

        free(data);
    }

    return 0;
}

sainath@Sainath:~/OSLAB/practical_8$ ./program2
Before fork: data[0] = 10
Child before modification: data[0] = 10
Child after modification: data[0] = 99
Parent after child modification: data[0] = 10
```

history:

```
490 nano program1.c
491 cd OSLAB
492 ls
493 cd practical_8
494 gedit program1.c
495 gcc program1.c -o program1
496 ./program1
497 gedit program2.c
498 gcc program2.c -o program2
499 ./program2
500 cat program1.c
501 ./program1
502 cat program2.c
503 ./program2
504 history
```